GEOSPATIAL AI · APPLIED RESEARCH

Map Scientist

Geospatial AI · Map Inference · Road Intelligence · Research-to-Production

📍 Tasco Building, Hanoi – Pham Hung | 🏢 TASCO Mapping | 👤 Individual Contributor | 📊 Reports to Head of Mapping

ABOUT THE COMPANY

About Tasco

VISION

A technology-enabled investment ecosystem, integrating Smart Mobility, Insurance, and Investments in essential services and innovation.

Corporate Information

Stock Ticker	HUT (HNX)
Established	1971
Subsidiaries & Affiliates	88
Employees	~10,000
Consolidated Sales	$1.4 billion
Total Assets	$1.98 billion
Website	tasco.com.vn

STRATEGIC PARTNERS

- Geely Auto
- Mitsui & Co.
- IFC – World Bank Group

BUSINESS ECOSYSTEM

MOBILITY	INSURANCE	INVESTMENT
Tasco Auto VETC Carpla VETC Go	Tasco Insurance Tasco Finance	DNP Water · Tasco Mall Ana Mandara Villas Six Senses Inochi

WHY TASCO

Our Mobility Ecosystem in Numbers

TASCO AUTO

40k+	180	Top 5	16	14.8%
New Cars Sold	Showrooms	Tax-paying Retailers	Car Brands	Market Share

VETC — ETC & MOBILITY

4.2m	~300	2m	650k	75%
Car Owners	Parking Lots	Transactions / Day	Car Repairs	ETC Market Share

0 Job Overview

POSITION	Map Scientist
FOCUS	Geospatial AI, Road Attribute Inference, Lane-Level Map Intelligence, Map Quality Science
DEPARTMENT	TASCO.COM.VN Mapping
LOCATION	Tasco Building Hanoi, Pham Hung
LEVEL	Individual Contributor
REPORTS TO	Head of Mapping

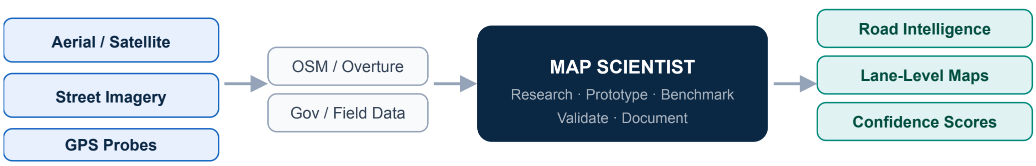
1 Job Purpose

The Map Scientist supports the development of AI-assisted map-making systems for Tasco Maps, MyTasco, VETC mobility services, navigation, HD Maps / ADAS Maps, and future vehicle intelligence products.

This role sits between Map Operations, Data Science, AI Engineering, and Product — translating mapping problems into measurable scientific tasks: dataset design, annotation schema, model benchmarking, geospatial feature engineering, error analysis, and prototype development.

It is an **applied research role** focused on turning aerial imagery, street-level imagery, GPS probes, OSM / Overture data, traffic-sign data, POI data, and user feedback into scalable map intelligence for Vietnam.

HOW THE ROLE WORKS — DATA TO MAP INTELLIGENCE



2 Key Responsibilities

A. Geospatial AI Research & Prototyping

→ Read, summarize, and reproduce selected papers related to road graph extraction, lane-level topology, road attribute inference, map-matching, traffic-sign detection, POI extraction, and map quality evaluation.

→ Build small-scale prototypes for map inference tasks using Python, PostGIS, PyTorch, computer vision, graph algorithms, and geospatial data processing.

- → Translate research ideas into production-oriented experiments with clear inputs, outputs, baselines, metrics, limitations, and rollout risks.
- → Maintain internal research notes, experiment logs, and technical memos for the Mapping team.

B. Road Intelligence & Lane-Level Map Inference

- → Support development of models and rules to infer road attributes from aerial imagery, street imagery, OSM, GPS probes, and field data.
- → Work on lane count estimation, road width inference, turn-lane detection, road-surface classification, intersection structure analysis, and traffic-signal / traffic-sign context extraction.
- → Assist in building lane-level and intersection-level datasets for Hanoi, Ho Chi Minh City, expressways, toll stations, parking zones, service roads, and high-priority corridors.
- → Evaluate model outputs against ground truth from MapOps, field collection, authoritative data, and manual QA.

C. Map Quality Science & Confidence Scoring

- → Design quantitative metrics for map data quality: completeness, freshness, positional accuracy, topology correctness, attribute accuracy, false-positive rate, false-negative rate, and operational impact.
- → Build confidence scores for map objects such as road segments, turn restrictions, lane attributes, traffic signs, POIs, entrances, geofences, and addresses.
- → Analyze disagreement between multiple data sources: OSM, Overture Maps, satellite imagery, street imagery, GPS traces, government data, user reports, and internal business data.
- → Prioritize MapOps work using risk-based scoring: high-traffic roads, high-fine-risk areas, navigation-critical intersections, and stale data regions.

D. Data Annotation, Benchmarking & Evaluation

- → Define annotation guidelines for AI map tasks: lane marking, stop line, turn arrow, traffic sign, road boundary, POI signboard, entrance, parking polygon, toll station, and road access point.
- → Build benchmark datasets with clear train / validation / test splits by city, road class, imagery source, and complexity level.
- → Establish baseline models and compare performance using precision, recall, F1, IoU, topology correctness, graph connectivity error, routing impact, and human-review cost.
- → Conduct systematic error analysis to identify whether errors come from data quality, annotation inconsistency, model design, imagery quality, road occlusion, GPS noise, or ambiguous legal rules.

E. AI-Assisted Map Operations

- → Build tools that help MapOps review model predictions faster: candidate generation, object ranking, auto-flagging, active learning queues, and QA dashboards.
- → Work with MapOps Specialists to convert AI predictions into validated map updates.
- → Create SOPs for human-in-the-loop validation of AI-generated map features.
- → Support a practical pipeline where AI proposes, MapOps verifies, Engineering publishes, and Product monitors user impact.

F. Data Sources, Fusion & Validation

- → Integrate and validate signals from OSM, Overture Maps, satellite imagery, street-level imagery, GPS probes, government data, field collection, user reports, and Tasco/VETC business data.
- → Design conflation logic to reconcile geometry, topology, attributes, source reliability, and temporal freshness across datasets.
- → Support source reliability scoring, data lineage, rollback evidence, and review workflows for critical map features.
- → Partner with Engineering to expose validated outputs to internal map editor tools, QA dashboards, search systems, routing graph generation, and release pipelines.

G. Documentation, SOPs & Team Enablement

- → Build documentation for dataset specs, annotation rules, benchmark design, model evaluation, confidence scoring, and release readiness.
- → Train junior MapOps members and contributors on AI-assisted validation and map quality standards.
- → Work with Product, Engineering, Data Science, and Business teams to shape the map intelligence roadmap.

3 Requirements

MUST-HAVE

- ✓ Bachelor's degree in GIS, Geomatics, CS, Data Science, EE, Transportation Engineering, Applied Mathematics, or related field.
- ✓ 0–2 years in geospatial data, ML, computer vision, map operations, remote sensing, navigation, logistics, or mobility data products.
- ✓ Strong Python skills for data processing, geospatial analysis, experiment scripting, and automation.
- ✓ Working knowledge of SQL and PostgreSQL/PostGIS.
- ✓ Familiarity with geospatial data formats: GeoJSON, Shapefile, GeoPackage, PBF, Parquet, raster tiles, vector tiles, GeoTIFF.
- ✓ Basic understanding of OpenStreetMap: nodes, ways, relations, tags, turn restrictions, road classes, POIs, admin boundaries.

NICE-TO-HAVE

- ✓ Experience with PyTorch, TensorFlow, scikit-learn, XGBoost, CatBoost, or similar ML frameworks.
- ✓ Computer vision: YOLO, U-Net, SegFormer, Mask2Former, SAM, OCR, or object detection / segmentation pipelines.
- ✓ Graph algorithms, road graphs, routing engines, map-matching, Valhalla, OSRM, GraphHopper, or NetworkX.
- ✓ Remote sensing, satellite imagery, aerial imagery, street-level imagery, dashcam data, or LiDAR point clouds.
- ✓ QGIS, JOSM, iD Editor, Overpass Turbo, osmium, osm2pgsql, DuckDB, GDAL, rasterio, or GeoPandas.
- ✓ Active learning, human-in-the-loop labeling, data labeling QA, or model-assisted annotation.
- ✓ OSM contribution history, GitHub portfolio, Kaggle / research project, or

- ✓ Basic ML understanding: supervised learning, model evaluation, train/test split, overfitting, precision/recall, confusion matrix, error analysis.
- ✓ Strong written communication in English.

thesis in mapping, CV, GIS, traffic, or mobility.

SOFT SKILLS & MINDSET

- ✓ Scientific mindset: skeptical, evidence-based, comfortable with uncertainty.
- ✓ Strong attention to data quality and edge cases.
- ✓ Able to turn ambiguous mapping problems into measurable experiments.
- ✓ Comfortable with messy real-world data: noisy GPS, occluded imagery, inconsistent labels.
- ✓ Practical orientation: reducing human review cost and improving product impact.
- ✓ Able to work independently, document assumptions, and ask precise technical questions.
- ✓ Interested in building Vietnam-localized map intelligence rather than generic global models.

4 Success Metrics — 6 to 12 Months

- Reproduce and document at least 2 relevant mapping / geospatial AI papers or open-source methods.
- Deliver at least 1 internal benchmark dataset for a priority problem (lane count, turn lane, traffic sign, POI signboard OCR, road width, or map-change detection).
- Build at least 1 working prototype that generates candidate map updates for MapOps review.
- Reduce manual review workload for one selected MapOps workflow through AI candidate ranking or active learning.
- Establish baseline metrics and error taxonomy for at least one map intelligence task.
- Create confidence scoring logic for at least one map object type: road attribute, POI, traffic sign, entrance, or geofence.
- Publish internal documentation: dataset spec, annotation guideline, model benchmark, error analysis, and next-step recommendation.
- Support Engineering / MapOps in moving one prototype toward a repeatable pipeline.

5 Product Scope

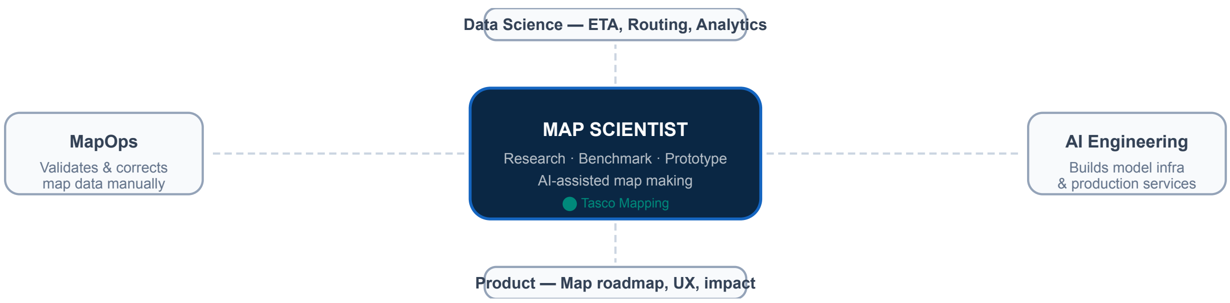
Product Line	Responsibility
V-Base-Maps	Road attributes, POI intelligence, map freshness, data completeness
V-Navigation-Maps	Lane guidance, turn restrictions, speed limits, road graph quality

HD Maps / ADAS Maps	Lane-level attributes, road geometry, intersection intelligence
Location & AI Search	POI confidence, signboard OCR, address / entrance signals
Contributor Platform	AI-assisted candidate generation, validation queues, QA scoring
Mobility Data Platform	GPS probe analysis, trajectory features, traffic and safety signals
Map Editor / Internal Tools	Human-in-the-loop review, model-assisted editing, QA dashboard

6 Technical Working Areas

- Geospatial AI benchmarks
- Satellite / aerial imagery
- Street-level imagery
- Road graph & lane topology
- Traffic sign / lane marking detection
- POI signboard OCR
- GPS probe feature engineering
- Map confidence scoring
- Active learning & HITL labeling
- PostGIS / DuckDB / Python
- Model evaluation dashboards
- Research-to-production documentation

7 How This Role Differs



Role	Main Ownership
Map Operations Specialist	Maintains and validates map data manually / operationally
POI / Geocoding Ops Specialist	Owns POI, address, search, geocoding, and geofence quality
Data Scientist	Builds analytics / ML models for ETA, routing quality, trajectory intelligence, and optimization
AI Engineer	Builds scalable model training and inference infrastructure
Map Scientist	Converts map problems into scientific experiments, benchmarks, prototypes, and AI-assisted map-making workflows

MapOps ensures the map is correct today. Engineering ensures the system scales. Data Science models trip and route behavior. ***Map Scientist*** *builds the scientific bridge that helps Tasco Maps infer, validate, and update map data faster and more intelligently.*

How to Apply

Send your email with subject **[Tasco Mapping Application - Map Scientist]** with your CV and a portfolio or description of relevant projects in GIS, mapping, computer vision, machine learning, remote sensing, road networks, OSM, traffic, mobility, or geospatial data science.

phung.dm@tasco.com.vn

tuyendung@vtii.vn

`[Tasco Mapping Application – Map Scientist]`

GitHub repositories, OSM contribution links, notebooks, thesis work, research summaries, benchmark reports, or sample geospatial analysis projects are encouraged.